

MID 2023

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Bangabandhu Sheikh Mujibur Rahman Science and Technology University
Department of Computer Science and Engineering

4th Year 1st Semester B.Sc. Engineering Mid-term Examination-2023

Course Code: CSE407

Course Title: Digital System Design

Total Marks: 20

Time: 1 (One) Hour

Answer all questions.

1. a) Design an accumulator that can perform the following microoperations. 8

$$p_1: A \leftarrow A + B$$

$$p_2: A \leftarrow A \vee B$$

$$p_3: A \leftarrow A \oplus B$$

- b) Determine the content of A after each microoperation if initially $A = 1100$ and $B = 1001$. 2

2. Design an arithmetic logic circuit with three selection variables S_1, S_0 and C_{in} that generates the following operations. Draw the logic diagram of one typical stage. 10

S_1	S_0	C_{in}	Function
0	0	0	$F = A + B$
0	0	1	$F = A$
0	1	0	$F = A + B + 1$
0	1	1	$F = A - 1$
1	0	0	$F = A + B' + 1$
1	0	1	$F = A + 1$
1	1	0	$F = A + B'$
1	1	1	$F = A$

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Section-A

1. a) Design an arithmetic unit that generates the following arithmetic operations. 8

S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$A = B'$	$B = A'$
0	1	$A = A + B$	$B = B + A$
1	0	$A = A + B' + 1$	$B = B + A' + 1$
1	1	$A = A' + B + 1$	$B = B' + A + 1$

- b) Show the hardware that implements the following statements. Include the logic gates for control function. [A , B and P are 4-bit registers] 2

$$T_1 + xy'T_2: A \leftarrow A + B, P \leftarrow P - 1$$

2. a) Using JK flip-flops design one typical stage of a register that performs the following logic microoperations: (draw necessary figures) 8

$$p_1: A \leftarrow A + 1$$

$$p_2: A \leftarrow A - 1$$

$$p_3: A \leftarrow A \oplus B$$

$$p_4: A \leftarrow A.B$$

- b) What are the components of a microprogram control? 2

3. a) Draw the circuit diagram for a status register containing following status bits- 4

Z – Zero [when the result is zero]

P – Parity [when no. of 1's in the result is odd]

V – Overflow [when the result contains extra bit]

N – Negative [when the result is negative]

- b) Draw the data formats of arithmetic and logical operands in memory and draw the instruction formats of memory-reference and register-reference instructions in memory. 3

- c) Discuss the differences between macrooperations and microoperations. 3

4. a) Illustrate the shift and adder multiplier architecture. 6

- b) Explain the read operation $A \leftarrow M[AR]$ and write operation $M[AR] \leftarrow A$ with address bus and data bus. 2

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Course Code : CSE405

Course Title : Digital System Design

Marks: 20
Time: 1 hour

N.B. Answer all 3(Five) questions, taking

Q1. Design a 4-bit combinational logic shifter with the following operations:

H_1	H_0	$H_2=0$	$H_2=1$
0	0	$F=A$	$F=A$
0	1	$F = \text{shr } A \text{ with input } I_R$	$F = \text{shr } A \text{ with LSB}$
1	0	$F = \text{shl } A \text{ with input } I_L$	$F = \text{shl } A \text{ with MSB}$
1	1	$F = \text{all } 0\text{'s}$	$F = \text{all } 1\text{'s}$

Q2. Design an arithmetic unit using two selection variables, S_1 and S_0 , that generates the following arithmetic operations (use an adder and basic gates)

S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A$	$F = A + 1$
0	1	$F = A - B - 1$	$F = A - B$
1	0	$F = B$	$F = B + 1$
1	1	$F = B - A - 1$	$F = B - A$

- Q3. Using JK flip-flops design one typical stage of a register that performs the following logical micro-operations: (draw necessary figures)

$$p_1: A \leftarrow \overline{A \vee B}$$

$$p_2: A \leftarrow \overline{A \wedge B}$$

$$p_3: A \leftarrow A \odot B$$

$$p_4: A \leftarrow A - B$$

write
write
in

Semester 2022

SECTION A (30 Marks)

1. (a) Design an arithmetic unit using two selection variables; S_1 and S_0 , that generates the following arithmetic operations (use an adder and basic gates).

8

S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A + B$	$F = A - B$
0	1	$F = A - 1$	$F = A$
1	0	$F = A \wedge B$	$F = A \vee B$
1	1	$F = \overline{A \wedge B}$	$F = \overline{A \vee B}$

- (b) What are the differences between macro-operations and micro-operations?

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